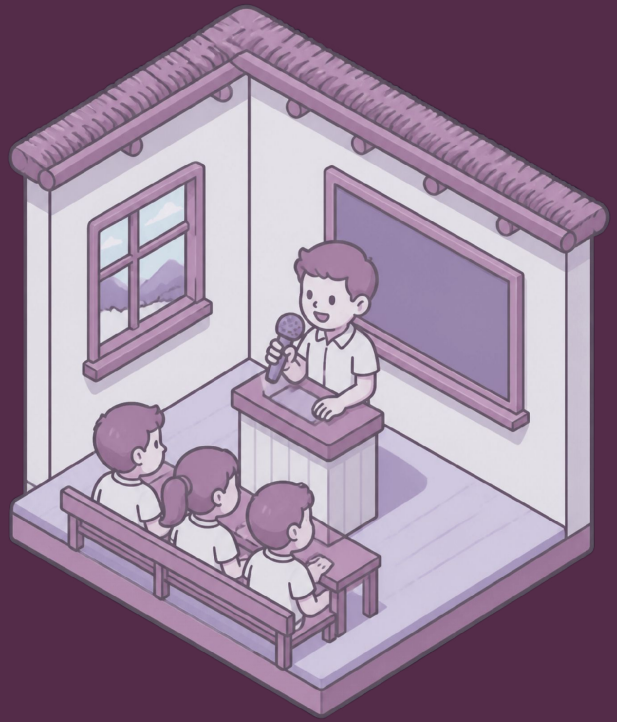
ENILS: Edge-Native Interactive Learning System

AI Tutoring That Works Without the Internet.

OFFLINE VOICE-FIRST LOW-COST



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The Problem: Why This Project?

No Connectivity

Unreliable or absent internet makes cloud-based AI tools completely unusable in rural and semi-urban government schools.

Overcrowded Classrooms

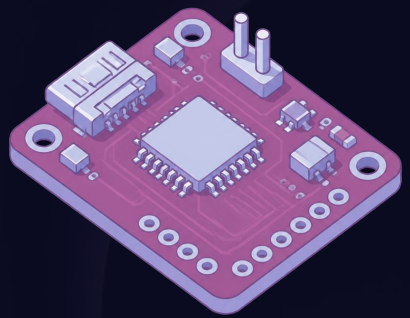
High teacher-to-student ratios leave little room for one-on-one doubt-clearing, especially in Math and Science.

Unaffordable Cloud APIs

Existing AI tutoring tools depend on paid cloud APIs — OpenAI, Google, Azure — unaffordable for donor-funded schools and useless without internet.

⚠ This creates a widening education gap based on geography and infrastructure — not ability.





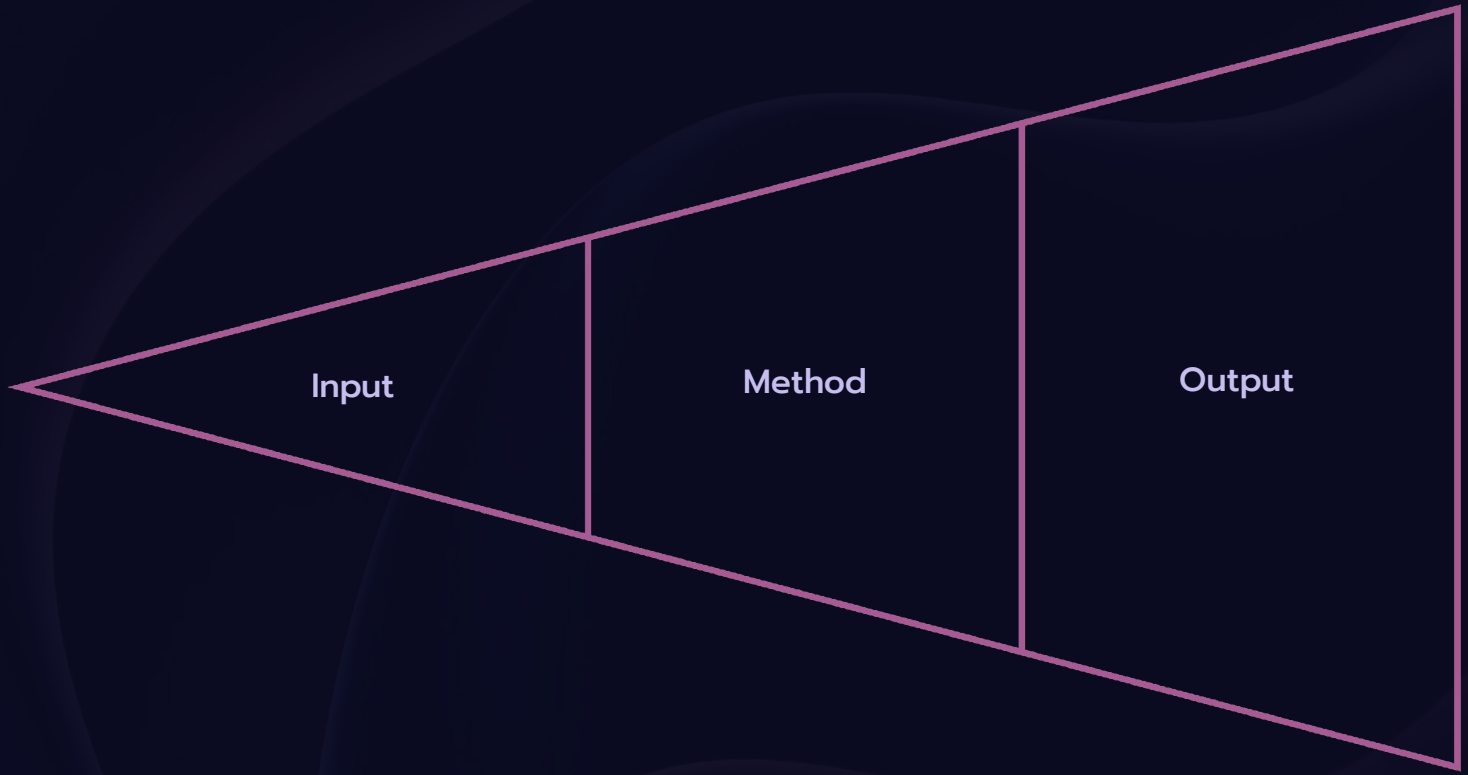
THE SOLUTION

What is ENILS?

A fully offline, voice-interactive AI tutoring device. A student speaks a question into a low-cost microphone — with **zero internet connection** — and the device listens, understands, thinks, and speaks back an age-appropriate explanation, like a patient human tutor.

Everything runs locally: speech recognition, the AI "brain," and voice response — on a laptop during development, and on a small low-power board (e.g., Raspberry Pi) once deployed.

How It Works: Input → Method → Output



The Python orchestration layer manages the full agent loop — Wake Word → STT → Prompt SLM → TTS Output — using async/threading and a simple state machine to keep responses smooth and crash-resistant.

Two Levels of Implementation

STAGE 1

Development — Laptop / PC

Build and debug the full voice pipeline with abundant compute resources.

- Intel i5 / Ryzen 5+ processor
- 8GB+ RAM — no GPU needed
- Easy debugging and iteration
- Full pipeline validated end-to-end

STAGE 2

Deployment — Edge Board (Raspberry Pi 4/5)

Port the working pipeline to a real, portable, classroom-usable device optimized for tighter resource limits.

- Low power consumption
- Compact and durable form factor
- Optimized for CPU/power/heat
- Ready for real classroom use

12-Week Implementation Timeline

1

Phase 1 — Weeks 1–4

Laptop MVP: Environment setup → local SLM via Ollama → offline STT integration → end-to-end CLI voice Q&A demo.

2

Phase 2 — Weeks 5–7

Logic & Integration: Subject-aware prompt engineering (Math/Science/Language) → wake-word detection → basic GUI showing conversation state.

3

Phase 3 — Weeks 8–10

Hardware Porting: OS setup on Raspberry Pi → cross-deploy model via llama.cpp → optimize for CPU, power, and heat.

4

Phase 4 — Weeks 11–12

Field Testing: Auto-start on boot, latency benchmarking → live classroom-style demo + full documentation.

Skills & Topics to Learn



Audio & Speech Processing

1. Mic input basics
2. Offline STT (Whisper)
3. Offline TTS (Piper/eSpeak-NG)
4. Noise & silence handling



Quantized SLMs — The Brain

1. LLM vs SLM concepts
2. Model quantization (GGUF, INT4/INT8)
3. Local model hosting (Ollama/llama.cpp)
4. Model selection trade-offs



Pipeline Orchestration

1. Building the agent loop in Python
2. Async/threading
3. State management
4. Error handling

Success Criteria & Outcomes

SUCCESS CRITERIA

What "Done" Looks Like

- Latency under **2.5 seconds**
- **100% offline** capability — zero internet dependency
- Reliable wake-word activation
- Accurate tutoring across Math, Science, and Language
- Stable **30+ minute** continuous run without crashing

Outcomes

What Gets Shipped

- Modular Python codebase (audio, STT, SLM, TTS, orchestration as separate modules)
- GitHub repo with README and setup guide
- System architecture flowchart + hardware diagram
- Benchmark report (laptop vs. Pi)
- Live demo/video with 3+ real tutoring interactions

Hardware & Budget

₹5–7.5K

Raspberry Pi 4/5

Recommended deployment board. Alternatives: Orange Pi 5 or Radxa Zero 3W for NPU acceleration.

₹1.5K

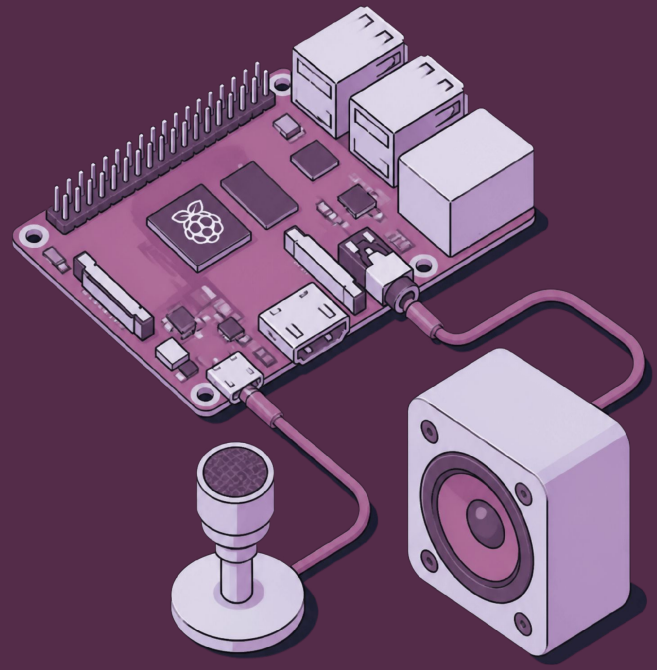
Peripherals

USB microphone (₹500–1,500), small speaker (₹300–800), microSD + case (₹500–800).

~$100

Total Per Unit

₹6,300–10,600 per device — affordable for school CSR grants or college project budgets.



ENILS: A Tutor for Every Student

ENILS brings always-on, personalized AI tutoring to classrooms that have never had reliable internet — turning a low-cost device into a patient, offline tutor for every student, regardless of geography or budget.

Always On

No internet. No cloud. No downtime.

Voice-First

Natural spoken interaction for every learner.

Truly Affordable

Under $125 per unit — built for real schools.



THANK YOU